Multimodal Prediction of Conversational Dynamics

Turn-Taking, Backchanneling, and Listening using Text, Audio, and Visual Signals

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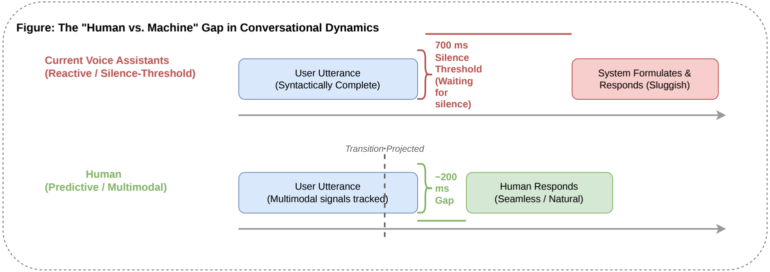
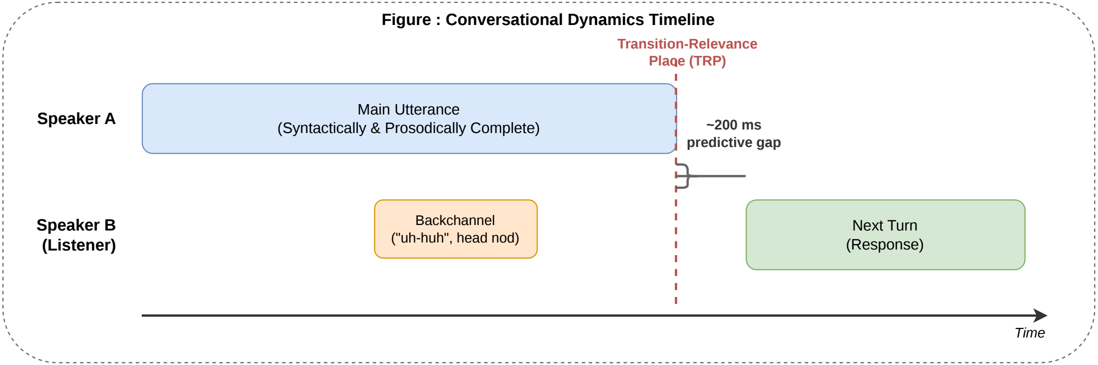
The "Human vs. Machine" Gap

- **Reactive Assistants**: Rely on silence thresholds (700ms).
- **Human Interaction**: Highly predictive; gaps are typically only 200ms.
- **The Goal**: Mimic human-like responsiveness by tracking multimodal signals in parallel.



figureReactive vs. Predictive systems

Understanding Turn-Taking and Backchanneling



- **Turn-Taking**: Handing over the "conversational floor."
- **Backchanneling**: Short overlapping signals (nods, "uh-huh") to show engagement without taking the floor.

Task Definition

Given a word utterance within an ongoing conversation, the model must assign one of three mutually exclusive labels:

- **KEEP**: The current speaker continues; the system remains silent.
- **TURN**: A turn shift occurs; the system takes the floor.
- **BACKCHANNEL (BC)**: The listener produces brief feedback (e.g., "mm-hmm") without taking the floor.

The Objective

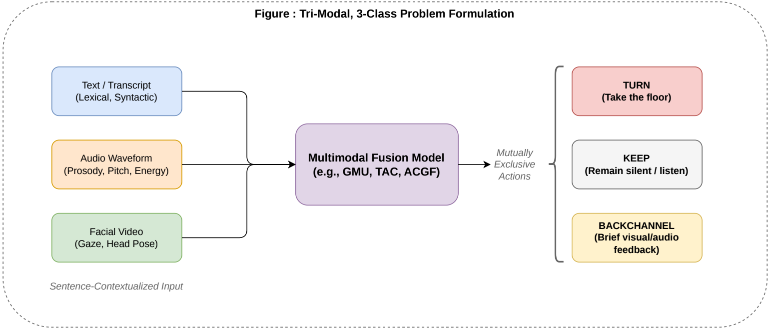
This is a **word-level**, **sentence-contextualized**, three-class classification problem. The model must integrate multimodal signals to project the optimal listener action in real-time.

Inputs:

- Text (GPT-2)
- Audio (HuBERT)
- Video (VideoMAE)

Target Actions:

- **KEEP**: Listener stays silent.
- **TURN**: Turn shift occurs.
- **BACKCHANNEL**: Brief feedback.



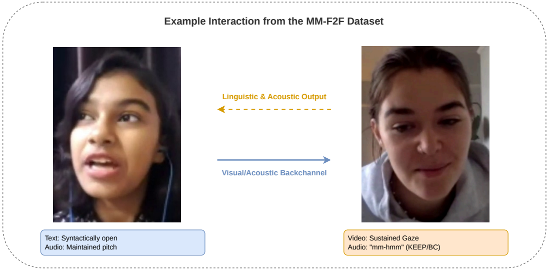
figureTri-modal prediction model.

Dataset: Multi-Modal Face-to-Face (MM-F2F)

Overview

The first large-scale face-to-face conversation dataset providing synchronized tri-modal data.

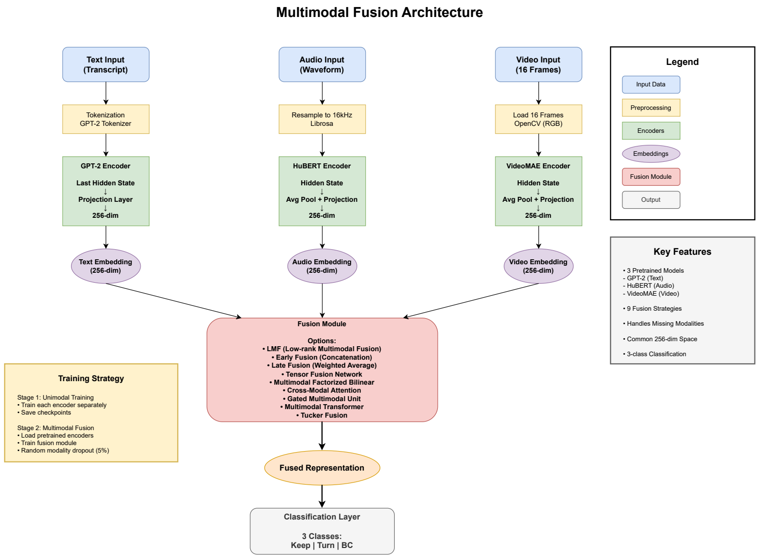
- **Scale**: 773 videos, ∼210 hours of conversation, and 955 unique speakers.
- **Annotations**: Over 169,000 word-level labels for TURN, BC, and KEEP.
- **Modalities**:
  - **Text**: Transcribed transcripts.
  - **Audio**: Raw waveforms (Prosody/Pitch).
  - **Video**: De-identified facial frames.



Representative Interaction

Figure 2.1: A dyadic interaction where the listener (right) decodes the speaker's (left) signals to project actions.

Two-Stage Training Framework



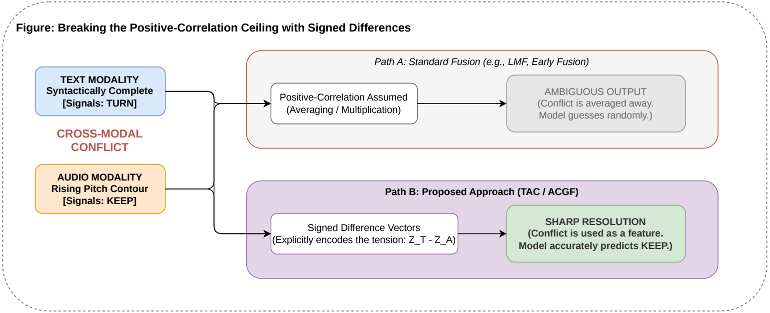
Training Strategy

- **Stage 1**: Train unimodal encoders separately (GPT-2, HuBERT, VideoMAE).
- **Stage 2**: Freeze all encoders; train only the **Fusion Module** on fixed features.

Key Benefit

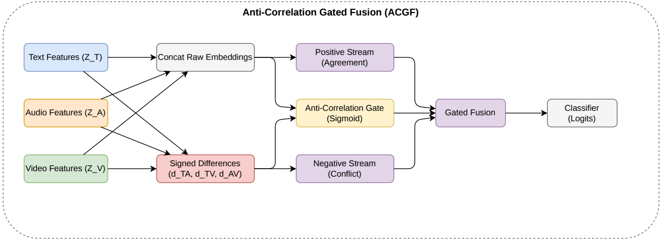
Isolates fusion quality from representational learning.

Breaking the Positive-Correlation Ceiling



- **The Ceiling**: Existing models (LMF, GMU) only amplify signals when modalities agree.
- **The Limitation**: Conflict (e.g., text says "stop" but audio says "continue") is averaged away.

Proposed: Anti-Correlation Gated Fusion (ACGF)



- Uses **Signed Difference Vectors** to capture directional discrepancy.
- **Dual Streams**: Separate pathways for Agreement and Conflict signals.
- **Learned Gating**: Dynamically decides which stream is more informative.

Proposed Model: Anti-Correlation Gated Fusion (ACGF)

Motivation: Hadamard-based fusions output ≈ 0 when two modalities have opposite signs, discarding the conflict entirely. ACGF explicitly models this gap.

Architecture (3 Components)

1. Signed Difference Vectors

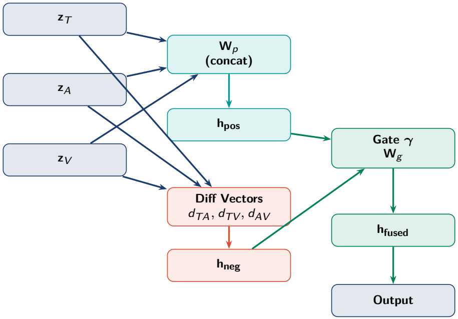
$$\mathbf{d}_{TA} = \mathbf{z}_T - \mathbf{z}_A, \quad \mathbf{d}_{TV} = \mathbf{z}_T - \mathbf{z}_V, \quad \mathbf{d}_{AV} = \mathbf{z}_A - \mathbf{z}_V$$

2. Dual Streams

$$\mathbf{h}_{\text{pos}} = \text{ReLU}(\mathbf{W}_p[\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V])$$
$$\mathbf{h}_{\text{neg}} = \text{ReLU}(\mathbf{W}_n[\mathbf{d}_{TA}; \mathbf{d}_{TV}; \mathbf{d}_{AV}])$$

3. Anti-Correlation Gate

$$\gamma = \sigma(\mathbf{W}_g[\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V; \mathbf{d}_{TA}; \mathbf{d}_{TV}; \mathbf{d}_{AV}])$$
$$\mathbf{h}_{\text{fused}} = \gamma \odot \mathbf{h}_{\text{pos}} + (1 - \gamma) \odot \mathbf{h}_{\text{neg}}$$



Params: ~788K Proposed

Deep Dive: How ACGF Captures Conflict

The Dual-Stream Logic

Standard fusions assume **Positive Correlation**:

- $\mathbf{z}_1 \odot \mathbf{z}_2$ is large only if signs agree.
- If signals oppose, the result collapses toward zero.

ACGF preserves the **Signed Difference**:

$$\mathbf{d}_{ij} = \mathbf{z}_i - \mathbf{z}_j$$

The magnitude encodes the *strength* of disagreement; the sign encodes the *direction* of conflict.

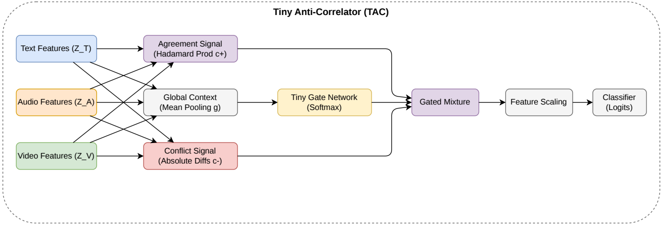
The Anti-Correlation Gate (γ)

The gate performs a **Dynamic Routing**:

$$\mathbf{h}_{\text{fused}} = \underbrace{\gamma \odot \mathbf{h}_{\text{pos}}}_{\text{Agreement}} + \underbrace{(\mathbf{1} - \gamma) \odot \mathbf{h}_{\text{neg}}}_{\text{Conflict}}$$

- **Agreement Dominates**: $\gamma \to 1$. Standard fusion behavior.
- **Conflict Dominates**: $\gamma \to 0$. The model extracts signals specifically from the *tension* between modalities.

Proposed: Tiny Anti-Correlator (TAC)



- **Goal**: Push the limits of parameter efficiency ($< 5,000$ params).
- Replaces linear projections with parameter-free element-wise math.
- High performance with a 159x reduction in parameters.

Proposed Model: Tiny Anti-Correlator (TAC)

Motivation: Can we distil the ACGF insight into < 5**K parameters** — small enough to run on edge devices?

Architecture (4 Steps)

1. Positive Correlation (Agreement)

$$\mathbf{c}^+ = \mathbf{z}_T \odot \mathbf{z}_A \odot \mathbf{z}_V$$

2. Negative Correlation (Conflict)

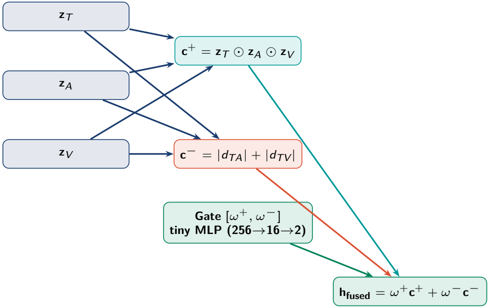
$$\mathbf{c}^- = |\mathbf{z}_T - \mathbf{z}_A| + |\mathbf{z}_T - \mathbf{z}_V|$$

3. Adaptive Gating (Global State)

$$\mathbf{g} = (\mathbf{z}_T + \mathbf{z}_A + \mathbf{z}_V)/3$$
$$[\omega^+, \omega^-] = \text{softmax}(\text{MLP}_{\text{tiny}}(\mathbf{g}))$$

4. Feature-Scaled Fusion

$$\mathbf{h}_{\text{fused}} = (\omega^+ \mathbf{c}^+ + \omega^- \mathbf{c}^-) \odot \boldsymbol{\lambda}$$
$$\hat{y} = W_{\text{cls}}\, \mathbf{h}_{\text{fused}}$$



Parameter Budget

Feature importance $\boldsymbol{\lambda}$	256
Tiny MLP (256→16→2)	4,354
Classifier (256 → 3)	771
Total	**5,173**

Deep Dive: The Mathematics of TAC Efficiency

Parameter-Free Feature Extraction

TAC eliminates heavy $\mathbf{W}_p$ and $\mathbf{W}_n$ matrices (∼400K params) via **Element-wise Inductive Biases**:

- **Agreement**: $\mathbf{c}^+ = \mathbf{z}_T \odot \mathbf{z}_A \odot \mathbf{z}_V$
- **Conflict**: $\mathbf{c}^- = |\mathbf{z}_T - \mathbf{z}_A| + |\mathbf{z}_T - \mathbf{z}_V|$

This forces the model to utilize the existing feature space without costly learned projections.

Diagonal Importance Scaling

Instead of a full 256×256 projection matrix, TAC uses a **Learnable Importance Vector $\boldsymbol{\lambda}$**:

$$\mathbf{h}_{\text{fused}} = (\omega^+ \mathbf{c}^+ + \omega^- \mathbf{c}^-) \odot \boldsymbol{\lambda}$$

- **Standard Layer**: 65,536 parameters.
- **TAC Scaling Vector**: **256 parameters**.
- **Impact**: Focuses only on scaling relevant inter-modal features at 1/256th the cost.

Tri-Modal Benchmark Results

Method	**Params**	**Acc**	**Mac-F1**	**BC-F1**
GMU (Baseline)	821,251	**0.8445**	**0.8607**	**0.9286**
ACGF (Proposed)	788,483	0.8409	0.8573	0.9231
LMF (Baseline)	37,027	0.8404	0.8558	0.9193
TAC (Proposed)	**5,173**	0.8378	0.8523	0.9111

- TAC performs within 0.84% of the much larger GMU model.
- Proves that fusion complexity is secondary to encoder quality.

Accuracy vs. Model Size

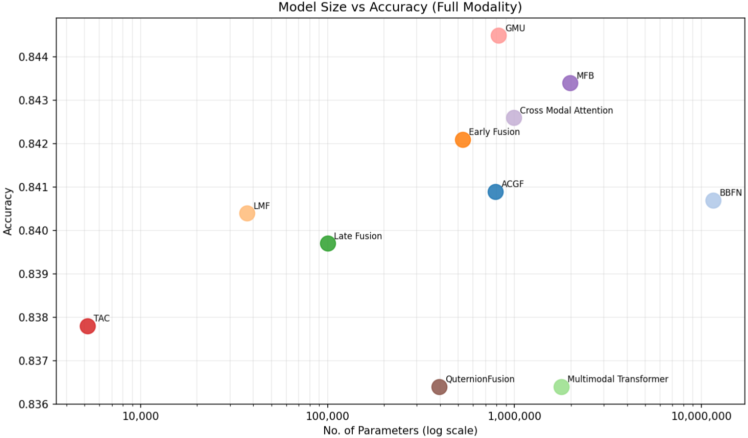


Figure 4.1: Log-scale analysis of fusion parameters vs. accuracy.

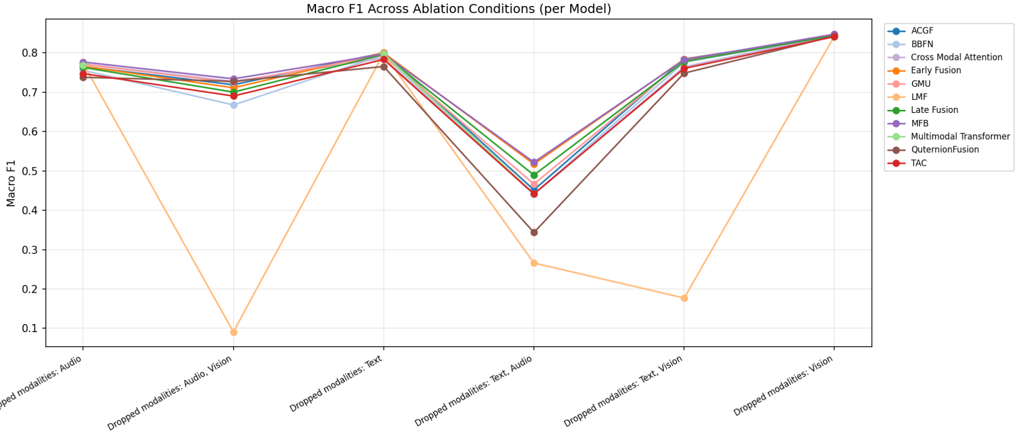
Decoupled Performance

- **No Correlation**: Increased parameter count does not guarantee higher accuracy.

Key Takeaway

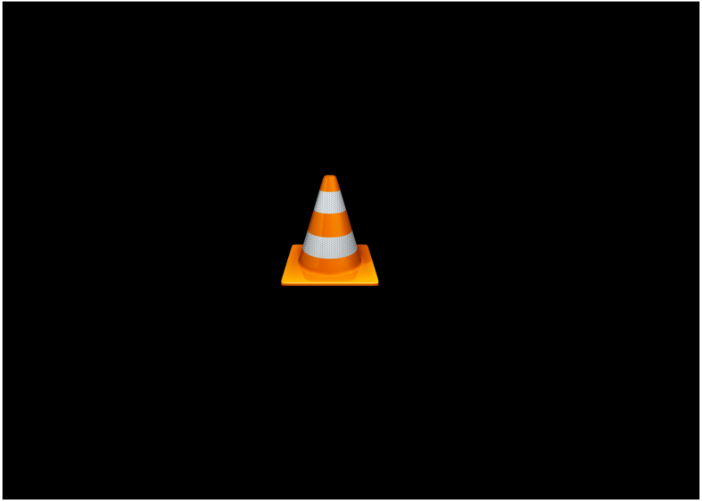
- **Ultra-Lightweight**: TAC (5K) achieves performance within 0.84% of GMU (821K).
- **Outperforms**: TAC beats larger models like MulT (1.78M).
- **TAC** and **ACGF** prove that the architectural approach to conflict is more vital than raw parameter scale.

Robustness Under Sensor Dropout



- **LMF**: Collapses below random chance when modalities are missing.
- **ACGF/TAC**: Maintain robustness by converting missing modalities into "conflict" signals.

Inference Demo: Input Signal Processing



Click image to play raw signal capture and feature extraction.

Processing Pipeline

The system processes parallel linguistic, acoustic, and visual streams to project Transition-Relevance Places (TRPs) and predict the conversational dynamics.

Raw Log / Signal Data

```
========================================
INPUT: Right
ACTION: keep [0.50117797 0.4973216  0.00150039]
========================================
========================================
INPUT: Right now,
ACTION: keep [0.6796622  0.29735368 0.02298414]
========================================
========================================
INPUT: Right now, the
ACTION: keep [9.8356700e-01 1.6220428e-02 2.1253760e-04]
========================================
========================================
INPUT: Right now, the problem
ACTION: keep [0.83811885 0.16080141 0.00107971]
========================================
========================================
INPUT: Right now, the problem is
ACTION: keep [0.907949   0.09034759 0.00170348]
========================================
========================================
INPUT: Right now, the problem is that
ACTION: keep [0.8629024  0.11558812 0.02150942]
========================================
========================================
INPUT: Right now, the problem is that I
ACTION: keep [0.96164286 0.03635665 0.00200053]
========================================
```

Key Contributions and Conclusion

- **Systematic Benchmark**: First comprehensive evaluation of 11+ fusion methods on MM-F2F.
- **Ultra-Efficiency**: Demonstrated that 5K parameters can rival 800K+ parameters.
- **Conflict Modeling**: Identified the "Positive-Correlation Ceiling" and proposed a way to transcend it.
- **Real-World Readiness**: TAC and ACGF offer deployable, robust solutions for full-duplex AI.

Thank You! Questions?

Appendix: Full Tri-Modal Benchmark Results

Table: Full tri-modal benchmark ranked by Macro-F1. † = proposed. **Bold** = best.

Rank	Method	Params	Acc	Mac-F1	Wt-F1	KEEP-F1	TURN-F1	BC-F1
1	**GMU**	821,251	**0.8445**	**0.8607**	**0.8441**	0.8145	0.8390	**0.9286**
2	MFB	1,976,067	0.8434	0.8588	0.8430	0.8139	**0.8390**	0.9235
3	Early Fusion	526,339	0.8421	0.8580	0.8419	**0.8156**	0.8348	0.9234
4	Cross-Attn	988,675	0.8426	0.8577	0.8418	0.8104	0.8395	0.9233
5	*ACGF*†	788,483	0.8409	0.8573	0.8411	0.8155	0.8332	0.9231
6	BBFN	11,579,911	0.8407	0.8565	0.8399	0.8065	0.8380	0.9249
7	LMF	37,027	0.8404	0.8558	0.8402	0.8134	0.8346	0.9193
8	Late Fusion	99,852	0.8397	0.8558	0.8397	0.8141	0.8321	0.9211
9	MulT	1,778,435	0.8364	0.8529	0.8364	0.8132	0.8259	0.9195
10	*TAC*†	5,173	0.8378	0.8523	0.8379	0.8135	0.8324	0.9111
11	Quaternion Fusion	395,011	0.8364	0.8482	0.8355	0.8074	0.8362	0.9010

Table: Fusion module parameter counts, ordered by scale.

Method	Category	Fusion Params
TAC (Proposed)	Anti-correlation	**5,173**
LMF (Liu et al.)	Tensor	37,027
Tucker	Tensor	99,852
Late Fusion	Decision-level	100,352
Quaternion Fusion	Algebraic	395,011
Early Fusion	Feature-level	526,339
TFN	Tensor	526,339
ACGF (Proposed)	Anti-correlation	**788,483**
GMU	Gating	821,251
Cross-Attn	Attention	988,675
MulT	Attention	1,778,435
MFB	Bilinear	1,976,067
BBFN	Gating	11,579,911

Appendix: Stage 1 Uni-modal Performance

Table: Performance of individual modality encoders. Bold = best per column.

Modality	Backbone	Mac-F1	Wt-F1	KEEP-F1	TURN-F1	BC-F1
Text	GPT-2	**0.751**	**0.747**	**0.767**	0.707	0.740
Audio	HuBERT	**0.751**	0.737	0.735	**0.805**	**0.759**
Video	VideoMAE	0.559	0.597	0.536	0.513	0.549

Observation

Text and Audio provide the strongest signals for turn prediction, while Video (VideoMAE) serves as a supplementary signal for nuance.

Appendix: Robustness to Missing Modalities

Table: Ablation study: Macro-F1 under sensor dropout conditions.

	GMU		LMF		ACGF†		TAC†	
Condition	Mac-F1	↓	Mac-F1	↓	Mac-F1	↓	Mac-F1	↓
Full Tri-modal	0.8607	—	0.8558	—	0.8573	—	0.8523	—
Text + Audio (−V)	0.8474	0.0133	0.8427	0.0131	0.8428	0.0145	0.8414	0.0109
Audio + Video (−T)	0.7943	0.0664	0.7985	0.0573	0.8003	0.0570	0.7835	0.0688
Text + Video (−A)	0.7710	0.0897	0.7733	0.0825	0.7654	0.0919	0.7475	0.1048
Text only (−A−V)	0.7249	0.1358	0.0909	0.7649	0.7191	0.1382	0.6904	0.1619
Audio only (−T−V)	0.7810	0.0797	0.1770	0.6788	0.7846	0.0727	0.7606	0.0917
Video only (−T−A)	0.4668	0.3939	0.2660	0.5898	0.4518	0.4055	0.4417	0.4106

- **LMF Vulnerability**: Tensor-based fusion collapses when inputs are zeroed out.
- **Proposed Advantage**: ACGF and TAC treat missing data as a "conflict" state, maintaining meaningful predictions.